

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

 show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 65

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

 That R3 reaches 192.168.2.0/24 through R1 only because R2's S0/0/1 was passive and not sending Hellos or LSAs. This confirms that passive interfaces suppress ospf adjacencies but still allow advertised routes to be learned from other routers.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

 R2(config)

 R2(config-router)

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

128 Accumulated cost metric: ~~128~~ ⁰ (Each serial link adds a cost of 64 so R3 to R2 to net is 128 total cost)

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To make sure that there is accurate cost differentiation among high speed interfaces. Because if left at 100 Mb/s, all interfaces greater than or equal to fast ethernet will appear as cost 1 which would mask path differences. Setting a higher reference can produce more realistic metrics.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

R1 to 192.168.3.0/24: Cost = Serial 781 + Gigabit 1 = 782

R1 to 192.168.23.0/30: 2 paths (both with same costs)

Path 1 Cost: R1 to R2 to R3 = 781 + 64 = 845

Path 2 Cost: r1 to R3 directly = 845

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

New accumulated cost: 1562. All serial interfaces now share the same bandwidth of 128 Kb/s after the bandwidth command, so each side advertises an equal cost, so 781 + 781 = 1562

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

__ This is because R1's direct serial link S0/0/1 was manually raised to cost 1565 which makes the indirect path through R2 (Cost = 1563) to be cheaper. Since ospf always selects the lowest cost path, traffic now shifts to R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

__ It is important because it ensures more predictable neighbor relationships and troubleshooting and prevents duplicate router IDs which break adjacency formations. Also, it is important for stable LSA database organization.

2. Why is the DR/BDR election process not a concern in this lab?

__ This is because all serial links are point to point and not multi access segments, so no DR/BDR election takes place. Also, the only broadcast LAN (g0/0) has a single router connected.

3. Why would you want to set an OSPF interface to passive?

__ To suppress any unnecessary Hello traffic on LANs with end hosts and to improve security while reducing CPU usage when advertising the network.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

__ Because of its multi area design where each area runs its own SPF which reduces LSA flooding scope.

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

____network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

____The route from EIGRP will be used because routing protocol selection is through administrative values and not simply metric values. EIGRP, OSPF, RIP will have administrative values of 90, 110, and 120 respectively, so EIGRP wins because it has the lowest administrative value.
